

Amazon EC2: Comprehensive Guide to Elastic Compute Cloud

Introduction

Amazon EC2 (Elastic Compute Cloud) is one of the most fundamental and widely used services in Amazon Web Services (AWS). It provides secure, resizable computing capacity in the cloud, allowing you to launch virtual servers (called instances) with the exact configuration you need. EC2 eliminates the need for upfront hardware investments, enabling you to scale computing resources up or down based on your application's demands.

This comprehensive document covers everything about Amazon EC2, from basic concepts to advanced features, use cases, pricing models, and best practices.

1. What is Amazon EC2?

Definition

Amazon EC2 is a web service that provides secure, resizable computing capacity in the AWS cloud. It allows you to launch virtual machines (instances) pre-configured with operating systems, applications, and computing resources that you can customize to meet your specific requirements.

Core Concept

Instead of purchasing, hosting, and maintaining physical servers, EC2 lets you:

- **Rent virtual computers on AWS's infrastructure**
- **Choose the operating system and software you need**
- **Scale capacity up or down instantly**
- **Pay only for the computing power you use**

Key Characteristics

Characteristic	Description
Elastic	Scale resources up or down based on demand
Virtual	Run fully virtualized server environments
Resizable	Change instance size as requirements evolve
Secure	Built on AWS security infrastructure with encryption and access controls
Flexible	Support for multiple operating systems and applications

2. EC2 Core Components

2.1 Instances

An instance is a virtual server in the AWS cloud. You can launch multiple instances with different configurations depending on your needs.

Instance Types:

- **On-Demand Instances:** Pay for compute capacity by the hour or second with no long-term commitments
- **Reserved Instances:** Make a one-time payment for a significant discount (up to 75%) compared to On-Demand pricing
- **Spot Instances:** Bid for unused EC2 capacity at up to 90% discount compared to On-Demand prices

- **Scheduled Instances:** Reserve capacity for specific time windows
- **Dedicated Instances:** Run on hardware dedicated to your use
- **Dedicated Hosts:** Full control over the physical server

2.2 Instance Types

EC2 offers hundreds of instance types optimized for different use cases:

General Purpose Instances

Best for: Web servers, small databases, development environments, microservices

Instance Family	vCPUs	Memory (GiB)	Use Case
t3/t3a	2	1-32	General purpose, cost-effective
t4g	2	0.5-32	General purpose, ARM-based
m5	2-72	8-288	Balanced compute, memory, network
m6i	2-72	8-288	General purpose, Intel-based
m6g	2-72	8-288	General purpose, ARM-based

Compute Optimized Instances

Best for: High-performance web servers, batch processing, gaming servers, scientific modeling

Instance Family	vCPUs	Memory (GiB)	Features
c5	2-72	4-144	High-performance processors
c6i	2-72	4-144	Intel-based compute
c6p	2-72	4-144	High-frequency Intel processors
c7g	4-32	8-128	ARM-based, high performance

Memory Optimized Instances

Best for: High-performance databases, real-time analytics, memory-intensive applications

Instance Family	vCPUs	Memory (GiB)	Special Features
r5	2-72	16-768	Balanced memory
r6i	2-72	16-768	Intel-based memory
r6g	2-72	16-768	ARM-based memory
x1	4-64	192-24,384	Massive memory, SAP HANA

m6g	2-72	8-288	Memory-intensive workloads
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Storage Optimized Instances

Best for: High-frequency databases, log processing, data analytics

Instance Family	vCPUs	Storage	Use Case
i3	8-56	Up to 250 TB NVMe	High-frequency transactional databases
i3en	8-56	Up to 720 TB NVMe	Large storage, high I/O
d2	8-36	Up to 14 TB HDD	Dense storage, data mining
h1	8-32	Up to 50 TB HDD	Large dataset storage

GPU Instances

Best for: Machine learning, graphics rendering, video encoding, scientific simulations

Instance Family	GPU	Use Case
p3	1-8 NVIDIA V100	ML training, high-performance computing

p4	1-8 NVIDIA A100	ML training, inference
g4	1-2 NVIDIA T4	ML inference, graphics
g5	1-8 NVIDIA A10G	ML inference, graphics rendering
vt1	3 NVIDIA Virtual	Video transcoding

2.3 Amazon EBS (Elastic Block Store)

EBS provides block-level storage volumes for use with EC2 instances.

EBS Volume Types:

Volume Type	Best For	Performance	Cost
General Purpose (gp3)	Most workloads	Up to 100,000 IOPS	Moderate
General Purpose (gp2)	Beginner workloads	Up to 16,000 IOPS	Lower
Provisioned IOPS (io2)	Critical database workloads	Up to 1,000,000 IOPS	Highest
Provisioned IOPS (io1)	I/O-intensive workloads	Up to 64,000 IOPS	High

Cold HDD (sc1)	Cold workloads	Lowest	Lowest
HDD (st1)	Frequently accessed data	Moderate	Low

Key EBS Features:

- **Persist data independently from instance life**
- **Snapshots for backup and replication**
- **Encryption support**
- **Auto-scaling storage capacity**

2.4 Amazon VPC (Virtual Private Cloud)

VPC lets you launch AWS resources in a virtual network that you define.

VPC Components:

- **Subnet: Segment of your VPC IP range**
- **Route Table: Defines where network traffic is directed**
- **Internet Gateway: Enables communication between VPC and internet**
- **Security Group: Virtual firewall controlling inbound/outbound traffic**
- **Network ACL: Controls traffic in and out of subnets**
- **VPN Connection: Secure connection to your network**

2.5 Security Groups

Security Groups act as virtual firewalls for EC2 instances, controlling inbound and outbound traffic.

Key Features:

- **Rule-based traffic control (allow/deny)**
- **Applied at instance level**
- **Support for TCP, UDP, and ICMP protocols**
- **Can reference other security groups**
- **Default security group allows all outbound traffic**

2.6 Elastic IPs

Elastic IP Addresses are static IPv4 addresses designed for dynamic cloud computing.

Features:

- **Remain allocated until you release them**
- **Can be mapped to any instance in your account**
- **Useful for maintaining consistent IP addresses**
- **One free Elastic IP per AWS account (per region)**

2.7 AMI (Amazon Machine Image)

AMI is a template containing the software configuration (operating system, application server, and applications) required to launch your instance.

AMI Types:

- **AWS Official: Provided by AWS (Amazon Linux, Ubuntu, Windows)**
 - **AWS Marketplace: Third-party images from vendors**
 - **Custom: Created by you from your own instances**
 - **Community: Shared by other AWS users**
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3. How EC2 Works

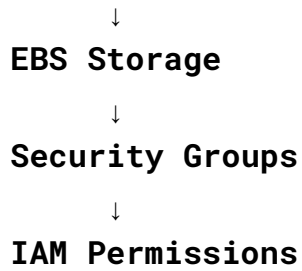
Launch Process

1. **Choose an AMI: Select operating system and pre-installed software**
2. **Select Instance Type: Choose compute, memory, and storage configuration**
3. **Configure Instance: Set network, security, and storage options**
4. **Add Tags: Label your instance for organization**
5. **Configure Security Group: Define firewall rules**
6. **Launch Instance: Create and start the virtual server**
7. **Connect: Access via SSH (Linux) or RDP (Windows)**

Architecture Flow

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User Request → EC2 Service → VPC Network → Instance Launch → OS Boot → Application Ready



Instance Lifecycle

Stage	Description
Pending	Instance is being created
Running	Instance is active and accessible
Stopping	Instance is shutting down
Stopped	Instance is off but resources retained
Shutting-down	Instance is being terminated
Terminated	Instance deleted, resources released

4. Key EC2 Features

4.1 Auto Scaling

Auto Scaling automatically adjusts the number of EC2 instances based on demand.

Benefits:

- **Maintain application performance during traffic spikes**
- **Reduce costs by scaling down during low usage**
- **Ensure availability with minimum instance count**
- **Automatic health check and replacement**

Auto Scaling Components:

- **Launch Template: Defines instance configuration**
- **Auto Scaling Group: Collection of instances**
- **Scaling Policies: Rules for adding/removing instances**
- **Scheduling: Pre-defined scaling schedules**

4.2 Elastic Load Balancing

ELB automatically distributes incoming application traffic across multiple EC2 instances.

Load Balancer Types:

Type	Best For	Protocol Support
Application Load Balancer (ALB)	HTTP/HTTPS applications	HTTP, HTTPS, WebSocket
Network Load Balancer (NLB)	High-performance needs	TCP, UDP, TLS
Gateway Load Balancer (GWLb)	Third-party security	Gateway Endpoints

Benefits:

- **Improved application availability**
- **Fault tolerance across instances**

- **Seamless scaling**
- **SSL/TLS termination**

4.3 Placement Groups

Placement Groups control how instances are distributed across physical hardware.

Types:

- **Cluster:** Instances in same low-latency node (for high-performance computing)
- **Spread:** Instances on distinct hardware (for critical applications)
- **Partition:** Instances in specific partitions (for distributed workloads)

4.4 Dedicated Instances and Hosts

Dedicated Instances: Run on hardware dedicated to your use (shared with other AWS customers in same family)

Dedicated Hosts: Full control over physical server (visible CPU, memory, storage)

Use Cases:

- **Compliance requirements**
- **License agreements requiring specific hardware**
- **Custom hardware configurations**

4.5 Instance Metadata and User Data

Instance Metadata: Information about your instance (IP address, instance ID, AMI ID)

User Data: Custom scripts or data passed to instance at launch (automated setup scripts)

4.6 EC2 Imbalanced Instances

Special instances with non-standard CPU-to-memory ratios for specific workloads.

4.7 EC2 Reserved Capacity

Reserved Instances: 1-year or 3-year commitment with significant discounts (up to 75%)

Capacity Reservations: Guarantee capacity in specific Availability Zone

5. EC2 Pricing Models

5.1 On-Demand Instances

- **Pay: By second (Linux) or hour (other OS) with no long-term commitment**
- **Discount: None (baseline pricing)**
- **Best For: Short-term, irregular workloads, testing, development**

5.2 Reserved Instances

Option	Commitment	Discount	Refundable
No Upfront	1 or 3 years	~40%	Yes
Partial Upfront	1 or 3 years	~50%	Partial
All Upfront	1 or 3 years	~75%	No

- **Best For: Predictable, steady-state workloads**

5.3 Spot Instances

- **Pay: Bid-based pricing for unused capacity**
- **Discount: Up to 90% compared to On-Demand**
- **Risk: Instance can be terminated if capacity needed**

- **Best For:** Fault-tolerant, flexible workloads, batch processing

5.4 Savings Plans

- **Commit:** Fixed hourly spend for 1 or 3 years
- **Discount:** Up to 72% compared to On-Demand
- **Flexibility:** Applies to any instance family, region, OS
- **Best For:** Steady usage with potential instance type changes

5.5 Dedicated Hosts

- **Pay:** Per host per hour
- **Best For:** Compliance, licensing, custom hardware needs

Pricing Comparison Example

Workload Type	On-Demand	Reserved (1yr)	Spot	Savings Plan
Monthly Cost (estimated)	\$100	\$50	\$10	\$30
Discount	0%	50%	90%	70%

6. Common EC2 Use Cases

6.1 Web Applications

Architecture:

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Users → Load Balancer → EC2 Instances (Web Servers) → RDS

Database → S3 (Static Assets)

Benefits:

- Scalable web server capacity
- Auto Scaling for traffic spikes
- High availability with multiple instances
- Cost optimization with Reserved Instances

6.2 Databases

Use Cases:

- MySQL, PostgreSQL, Oracle, SQL Server
- MongoDB, Cassandra, Redis
- Enterprise databases (SAP, IBM DB2)

Configuration:

- Provisioned IOPS EBS volumes for performance
- Multi-AZ deployment for availability
- Reserved Instances for steady capacity

6.3 Development and Testing

Benefits:

- Rapid environment creation
- Cost-effective with Spot Instances
- Easy teardown and recreation
- Multiple OS support

Best Practices:

- Use tagging for organization
- Implement automated shutdown schedules
- Use cost allocation tags

6.4 Enterprise Applications

Examples:

- ERP systems (SAP, Oracle)
- CRM platforms (Salesforce)
- Collaboration tools

Requirements:

- Dedicated Hosts for licensing
- High-memory instances

- **Persistent storage with EBS**

6.5 Big Data and Analytics

Use Cases:

- **Data processing pipelines**
- **Real-time analytics**
- **Machine learning training**

Configuration:

- **Compute-optimized instances (c5, c6i)**
- **Storage-optimized instances (i3, i3en)**
- **GPU instances for ML (p3, p4, g5)**